

one's class, which varies within (school,grade,year) cells.

30. **Chaney et al. (2012), The Collateral Channel: How Real Estate Shocks Affect Corporate Investment.** *Table 5.*

In regression Equation (1), the dependent variable is the value of investment in firm i and year t divided by the lagged book value of properties, plants, and equipments (PPE), and the explanatory variables are firm and time fixed effects and the market value of firm i in year t divided by its lagged PPE, and some controls. This regression corresponds to Regression 1, with some controls. The stable groups assumption is presumably satisfied: it is likely that between each pair of consecutive years, there are firms whose market value divided by their lagged PPE does not change. This is a sharp design.

31. **Aaronson et al. (2012), The Spending and Debt Response to Minimum Wage Hikes.** *Tables 1, 2, and 5.*

In regression Equation (1), the outcome variable is, say, income of household i at period t , and the explanatory variables are household and time fixed effects, and the minimum wage in the state where household i lives in period t . This regression corresponds to Regression 1. The stable groups assumption is satisfied: between each pair of consecutive periods, there are states where the minimum wage does not change. This is a sharp design.

32. **Brambilla et al. (2012), Exports, Export Destinations, and Skills.** *Table 5.*

In the regression in, say, the first column of Table 2, the dependent variable is a measure of skills in the labor force employed by firm i in industry j at period t , and the explanatory variables are firm and industry $\times$ period fixed effects, the ratio of exports to sales in firm i at period t , and some controls. This regression corresponds to Regression 1, with some controls. The stable groups assumption is presumably satisfied: it is likely that between each pair of consecutive periods, there are firms whose ratio of exports to sales does not change. This is a sharp design.

33. **Faye and Niehaus (2012), Political Aid Cycles.** *Table 3, Columns 4 and 5, and Tables 4 and 5.*

In regression Equation (2), the dependent variable is the amount of donations received by receiver r from donor d in year t , and the explanatory variables are donor $\times$ receiver fixed effects, an indicator for whether there is an election in country r in year t , a measure of alignment between the ruling political parties in countries r and d at t , and the interaction of the election indicator and the measure of alignment. This regression corresponds to Regression 1. The stable groups assumption is presumably not satisfied: it is unlikely that there are donor-receiver pairs that are perfectly unaligned. This is a sharp design.

7 Proofs

Theorem S1 relies on the following lemma.

Lemma S1 *If Assumptions 1 and 3-5 hold, for all $(g, g', t, t') \in \{1, \dots, G\}^2 \times \{1, \dots, T\}^2$,*

$$\begin{aligned} & E(Y_{g,t}|\mathbf{D}) - E(Y_{g,t'}|\mathbf{D}) - (E(Y_{g',t}|\mathbf{D}) - E(Y_{g',t'}|\mathbf{D})) \\ &= D_{g,t}E(\Delta_{g,t}^{TR}|\mathbf{D}) - D_{g,t'}E(\Delta_{g,t'}^{TR}|\mathbf{D}) - (D_{g',t}E(\Delta_{g',t}^{TR}|\mathbf{D}) - D_{g',t'}E(\Delta_{g',t'}^{TR}|\mathbf{D})). \end{aligned}$$

Proof of Lemma S1

For all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$,

$$\begin{aligned} E(Y_{g,t}|\mathbf{D}) &= E\left(\frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} (Y_{i,g,t}(0) + D_{i,g,t}(Y_{i,g,t}(1) - Y_{i,g,t}(0))) \middle| \mathbf{D}\right) \\ &= E(Y_{g,t}(0)|\mathbf{D}) + D_{g,t}E(\Delta_{g,t}^{TR}|\mathbf{D}). \end{aligned}$$

The end of the proof is the same as that of Lemma 1.

Proof of Theorem S1

The proof of that result is very similar to the proof of Theorem 1.

$$\begin{aligned} \beta_{fe} &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} Y_{g,t}}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right) \\ &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} E(Y_{g,t}|\mathbf{D})}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right) \\ &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} (E(Y_{g,t}|\mathbf{D}) - E(Y_{g,1}|\mathbf{D}) - E(Y_{1,t}|\mathbf{D}) + E(Y_{1,1}|\mathbf{D}))}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right) \\ &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} (D_{g,t}E(\Delta_{g,t}^{TR}|\mathbf{D}) - D_{g,1}E(\Delta_{g,1}^{TR}|\mathbf{D}) - D_{1,t}E(\Delta_{1,t}^{TR}|\mathbf{D}) + D_{1,1}E(\Delta_{1,1}^{TR}|\mathbf{D}))}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right) \\ &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t} E(\Delta_{g,t}^{TR}|\mathbf{D})}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right) \\ &= E\left(\frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t} \Delta_{g,t}^{TR}}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}\right). \end{aligned}$$

The second equality follows from the law of iterated expectations. The third and fifth equalities follow from Equations (5) and (6). The fourth equality follows from Lemma S1. The last equality follows from the law of iterated expectations.

Proof of Proposition S1

Assuming that $N_{g,2}/N_{g,1}$ does not vary across g ensures that there exists a strictly positive real number ϕ such that $N_{g,2}/N_{g,1} = \phi$. Then,

$$\begin{aligned}
\varepsilon_{g,2} &= D_{g,2} - D_{g,\cdot} - D_{\cdot,2} + D_{\cdot,\cdot} \\
&= D_{g,2} - \left(\frac{N_{g,1}}{N_{g,\cdot}} D_{g,1} + \frac{N_{g,2}}{N_{g,\cdot}} D_{g,2} \right) - D_{\cdot,2} + \left(\frac{N_{\cdot,1}}{N} D_{\cdot,1} + \frac{N_{\cdot,2}}{N} D_{\cdot,2} \right) \\
&= D_{g,2} - \left(\frac{1}{1+\phi} D_{g,1} + \frac{\phi}{1+\phi} D_{g,2} \right) - D_{\cdot,2} + \left(\frac{1}{1+\phi} D_{\cdot,1} + \frac{\phi}{1+\phi} D_{\cdot,2} \right) \\
&= \frac{1}{1+\phi} (D_{g,2} - D_{g,1} - D_{\cdot,2} + D_{\cdot,1}), \tag{25}
\end{aligned}$$

where the first and third equalities follow from the fact $N_{g,2}/N_{g,1}$ does not vary across g .

Then, the definition of $w_{g,2}^{TR}$, Equation (25) and Assumption 1 imply that

$$\begin{aligned}
w_{g,2}^{TR} &= \frac{(D_{g,2} - D_{\cdot,2})}{\sum_{g=1}^G \frac{N_{g,2}}{N_1} (D_{g,2} - D_{\cdot,2}) D_{g,2}} \\
&= \frac{(D_{g,2} - D_{\cdot,2})}{\sum_{g=1}^G \frac{N_{g,2}}{N_1} (D_{g,2} - D_{\cdot,2})^2}.
\end{aligned}$$

Proof of Theorem S2

The proof relies on the lemma below, which we start by proving, before proving the theorem.

Lemma S2 *If Assumptions 1-5 and S2 hold,*

$$\begin{aligned}
&E(Y_{g,t}|\mathbf{D}) - E(Y_{g,t-1}|\mathbf{D}) - (E(Y_{g',t}|\mathbf{D}) - E(Y_{g',t-1}|\mathbf{D})) \\
&= (D_{g,t} - D_{g,t-1})E(\Delta_{g,t}|\mathbf{D}) - (D_{g',t} - D_{g',t-1})E(\Delta_{g',t}|\mathbf{D}).
\end{aligned}$$

Proof of Lemma S2

By Lemma 1 and Assumption S2,

$$\begin{aligned}
&E(Y_{g,t}|\mathbf{D}) - E(Y_{g,t-1}|\mathbf{D}) - (E(Y_{g',t}|\mathbf{D}) - E(Y_{g',t-1}|\mathbf{D})) \\
&= D_{g,t}E(\Delta_{g,t}|\mathbf{D}) - D_{g,t-1}E(\Delta_{g,t-1}|\mathbf{D}) - D_{g',t}E(\Delta_{g',t}|\mathbf{D}) + D_{g',t-1}E(\Delta_{g',t-1}|\mathbf{D}) \\
&= (D_{g,t} - D_{g,t-1})E(\Delta_{g,t}|\mathbf{D}) - (D_{g',t} - D_{g',t-1})E(\Delta_{g',t}|\mathbf{D}).
\end{aligned}$$

Proof of the decomposition for the fixed-effect regression

First, we have

$$\begin{aligned}
& \sum_{g,t} N_{g,t} \varepsilon_{g,t} E(Y_{g,t} | \mathbf{D}) \\
&= \sum_{g,t} N_{g,t} \varepsilon_{g,t} [E(Y_{g,t} | \mathbf{D}) - E(Y_{1,t} | \mathbf{D})] \\
&= \sum_g \sum_{t=2}^T \left[\sum_{t' \geq t} N_{g,t'} \varepsilon_{g,t'} \right] [E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D}) - (E(Y_{1,t} | \mathbf{D}) - E(Y_{1,t-1} | \mathbf{D}))] \\
&= \sum_g \sum_{t=2}^T \left[\sum_{t' \geq t} N_{g,t'} \varepsilon_{g,t'} \right] [(D_{g,t} - D_{g,t-1}) E(\Delta_{g,t} | \mathbf{D}) - (D_{1,t} - D_{1,t-1}) E(\Delta_{1,t} | \mathbf{D})] \\
&= \sum_{(g,t): D_{g,t} \neq D_{g,t-1}, t \geq 2} \left[N_{g,t} (D_{g,t} - D_{g,t-1}) \sum_{t' \geq t} \frac{N_{g,t'}}{N_{g,t}} \varepsilon_{g,t'} \right] E(\Delta_{g,t} | \mathbf{D}). \tag{26}
\end{aligned}$$

The first equality follows by (6). The second equality follows from summation by part and (6). The third equality follows from Lemma S2. The fourth equality stems from the fact that by (6), the terms with $g = 1$ vanish.

Similarly,

$$\begin{aligned}
\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t} &= \sum_g \sum_{t=2}^T \left[\sum_{t' \geq t} N_{g,t'} \varepsilon_{g,t'} \right] [D_{g,t} - D_{g,t-1}] \\
&= \sum_{(g,t): D_{g,t} \neq D_{g,t-1}, t \geq 2} N_{g,t} (D_{g,t} - D_{g,t-1}) \sum_{t' \geq t} \frac{N_{g,t'}}{N_{g,t}} \varepsilon_{g,t'}. \tag{27}
\end{aligned}$$

The result follows by combining (4), (26), (27), and the law of iterated expectations.

Proof of the decomposition for the first-difference regression

First, we have

$$\begin{aligned}
& \sum_{(g,t): t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D})) \\
&= \sum_{(g,t): t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D}) - (E(Y_{1,t} | \mathbf{D}) - E(Y_{1,t-1} | \mathbf{D}))) \\
&= \sum_{(g,t): t \geq 2} N_{g,t} \varepsilon_{fd,g,t} [(D_{g,t} - D_{g,t-1}) E(\Delta_{g,t} | \mathbf{D}) - (D_{1,t} - D_{1,t-1}) E(\Delta_{1,t} | \mathbf{D})] \\
&= \sum_{(g,t): t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (D_{g,t} - D_{g,t-1}) E(\Delta_{g,t} | \mathbf{D}).
\end{aligned}$$

The first equality follows from (14). The second equality follows from Lemma S2. The third equality follows from (14) again. The result follows by combining (13) with the last display, and using the law of iterated expectations.